SZÉCHENYI ISTVÁN EGYETEM

Simulation and Experimental Correlation of the Audi 1.8T 20V Engine

Performance Analysis, Validation, and Recommendations

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1. Initial Engine Simulation

1.1 Motivation and Simulation Environment

The goal of this project is to build a 0D simulation model of the Audi/VW 1.8 20V engine, validate it against experimental data from a vehicle of the same configuration measured on a chassis dyno, and iteratively refine the model until simulation and measurement agree to within the target tolerance. A model correlated to this level becomes a reliable basis for predicting the effect of further design and calibration changes without returning to the test cell each time.

As technology advances, our options for carrying out different simulations of internal combustion engines continue to grow. Simulation provides data and results faster, more cost-effectively and more simply than real-world tests or physical measurements. The AVL Cruise M is a multidisciplinary vehicle system simulation tool for mobility concept analysis, subsystem design and layout, and virtual component integration [1.].

1.2 The VAG 1.8T 20V Engine

The 1.8T 20V is a well-suited subject for this study for several reasons. First, it is extensively documented: factory specifications, bore/stroke/rod dimensions and compression data are consistently published across technical and tuning references, which makes sourcing realistic model inputs straightforward. Second, it is widely regarded as a strong tuning platform — being turbocharged, it responds readily to boost and calibration changes, with a remap on a standard engine typically yielding around a 40–50 bhp increase, and the cast-iron block being solidly constructed enough to handle substantial power above the factory output. This means the relationship between a few key variables and the engine's output is both strong and well-characterised, which is exactly what makes it a good candidate for correlating a simulation against measured results [2.][3.].

Parameter	Value
Configuration	Inline 4-cylinder, turbocharged
Fuel system	Multi-point (port) injection, Gasoline
Valvetrain	20-valve (5/cylinder:3 intake,2 exhaust)
Displacement	~1,781 cm³
Bore	81.0 mm
Stroke	86.4 mm
Connecting rod length	144 mm
Compression ratio	~9.3:1
Aspiration	Turbocharged, intercooled
Engine code	AUQ

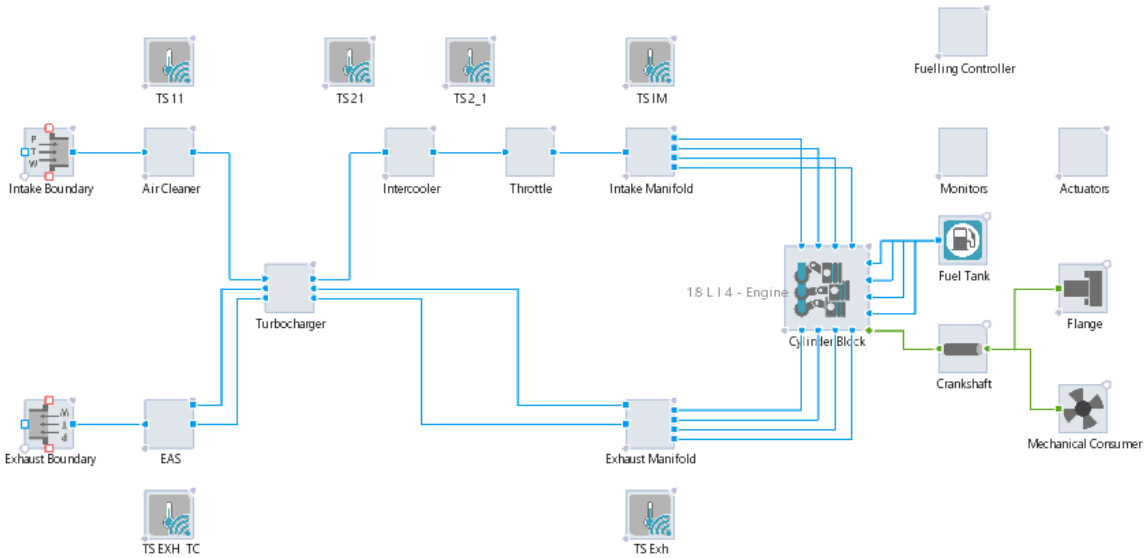
1. Table: Engine Parameters [4.]

The only modifications made to this engine are the following:

- RS3 Coils
- 500*300*76 TurboWorks cooler
- Sport Catalysator
- 76 mm exhaust system

1.3 Base Simulation Model

The base engine model was generated using Cruise M's Engine Generator, a wizard that assembles a complete, runnable engine layout from a relatively small set of fundamental inputs and a chosen architecture.

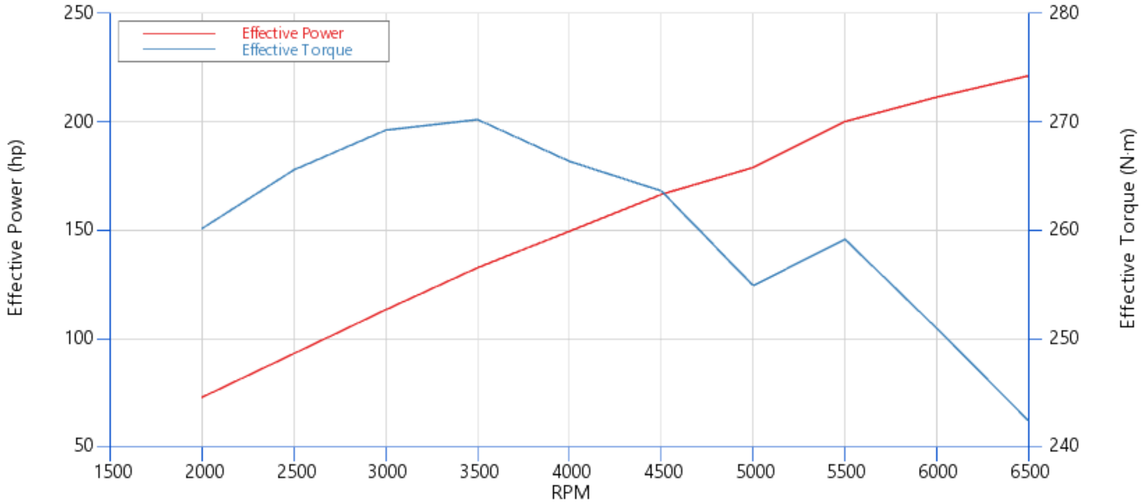


1. Figure: Base engine topology

The basic inputs required were:

- Engine geometry — bore, stroke, compression ratio, number of cylinders, engine configuration,
- Fuel type – Gasoline or Diesel
- Injection type – Direct or Port injection
- Charging type – NA or Turbocharged
- Cycle type – 4 or 2 stroke
- Engine speed
- Model type – 0D or 1D
- Valvetrain definition — number of valves per cylinder (the simulation can't run the 3 intake and 2 exhaust valve system, so it must be upscaled)

The generator populates each of these components with default template values, which produces a model that runs immediately but is only geometrically representative — the defaults are not the real component dimensions. Refining those defaults toward the actual hardware is what turns the demo-style layout into a meaningful representation of the 1.8T 20V.



2. Figure: Base Engine performance

The simulation results show that the base model produces 270 Nm effective torque and around 210 hp effective power. These values show higher performance numbers than the factory results, so I must adapt the simulation according to this.

1.4 Refined Simulation Model

After analysing the base results, I have decided to go through the whole topology and validate whether the wizard's results are correct. I have found that the following parametrizations were sufficient for my simulation:

- intake boundary, exhaust boundary, intake manifold
- the 6 temperature sensors, injectors, actuators, monitors,
- fuel tank, crankshaft, flange and mechanical consumer

So, I didn't change any possible parameters here.

On the other hand, the following changes were made to improve the simulation:

Component	Changed parameter	Base	Modified	Ref.
Air cleaner	Core length (mm)	827	32	[5.]
Turbocharger	Wastegate ø (mm)	49	25	-
Throttle body	ø (mm)	49	60	[6.][7.]
Throttle body	Shaft diameter (mm)	9.8	6	[6.][7.]
Intercooler	Flow length (mm)	1786	1000	[8.]
Exhaust manifold	Volume (L)	4.45	0.8	[9.]
Exhaust manifold	Reference area (mm²)	1963	1100	[9.]
Catalyst	Hydraulic ø (mm)	4.25	1.25	-
Catalyst	Reference volume (L)	7.92	2.0	-
Catalyst	Reference ø (mm)	56.87	76	-
Catalyst	Target pressure drop (bar)	0,008	0.012	-

2. Table: System refinement table

For the turbocharger wastegate refinement I couldn't find any exact parameters, rather than that I have measured the pictures that I have found on the internet and gave an estimate. The car has an aftermarket catalysator, which is not defined, so for that I had to use averages from different sport catalyst systems. The wastegate flow, defined as the ratio of turbine-to-total mass flow, is varied to achieve the compressor's target pressure ratio. The turbine is modelled as a fixed-size component, while the wastegate opens and closes to regulate boost pressure to the target value. This approach most closely represents the physical behaviour of the real engine: a fixed-geometry K03s turbocharger with a wastegate modulating to maintain the target boost. At high engine speeds, when the turbocharger reaches its flow limit, the wastegate is fully closed and boost tapers off naturally, as the turbine is unable to supply any additional flow.

The component requiring the most parameter changes was the cylinder block. The port injector was left at its default configuration. For future work, it would be worth adding an injection profile table, which could further optimise the simulation. For the intake and exhaust port I had to change a lot of variables. First I tried to add my own Valve lift table and with that I could have added more data about the valves to the simulation, but unfortunately it didn't work, so I had to go back to the basic setup and change the data with the information I had from catcam [10.].

Component	Changed parameter	Stock	Modified
Intake Port	Valve opening (deg)	346	342
	Valve closing(deg)	585	568
	Maximum Valve lift (mm)	7.2	7,6
	Number of valves (-)	2	2
	Flow coefficient multiplier (-)	0.777	1.165
Exhaust Port	Valve opening (deg)	138	150
	Valve closing(deg)	368	370
	Maximum Valve lift (mm)	6.8	9.2
	Number of valves (-)	2	2
	Flow coefficient multiplier (-)	0.582	0,582

3. Table: Valve lift refinement

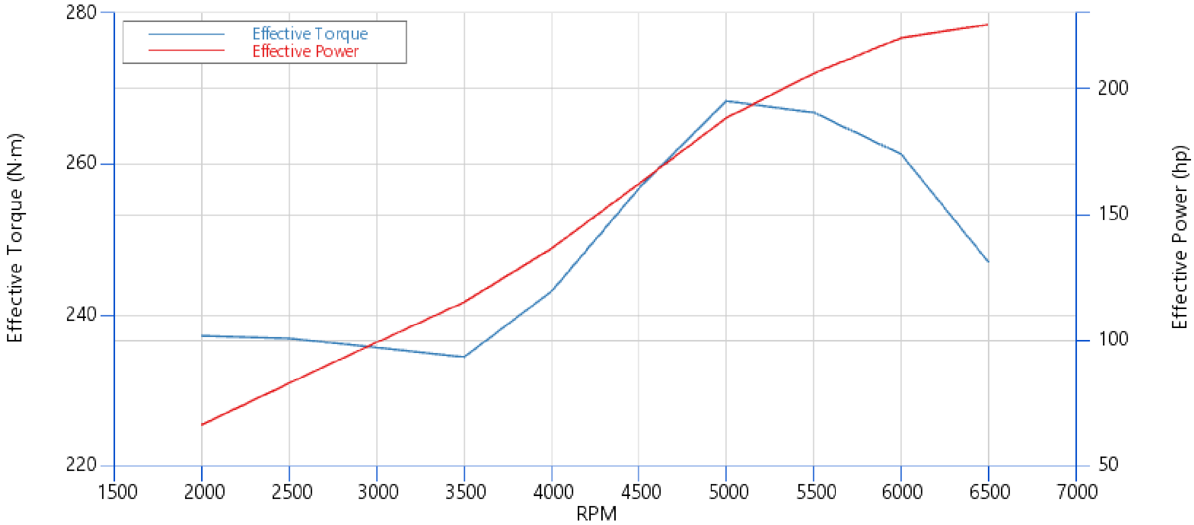
Another important factor, while setting these parameters up was the traces. For that I have (-1/2, +1/2) output interval and 2 engine cycles, because with these settings the results are clear, and it can be seen easily.

In the combustion chamber, the connecting-rod length was corrected from 141 mm to 144 mm, and a 1 mm piston pin offset was applied (from 0) [11.].

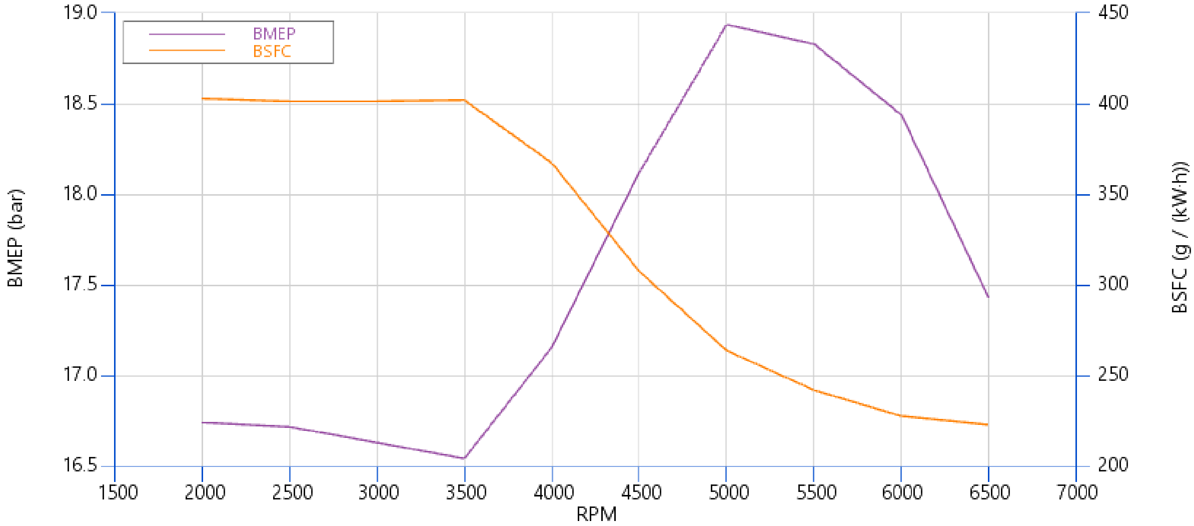
The other important change was setting the initial air–fuel ratio to 11.5 rather than 14.5, since the engine is being simulated at full load, where a turbocharged engine runs rich to control combustion temperatures and protect against knock — making ~11.5 the realistic operating point.

I have not changed the basic -18° start of combustion nor the 64.29 combustion duration.

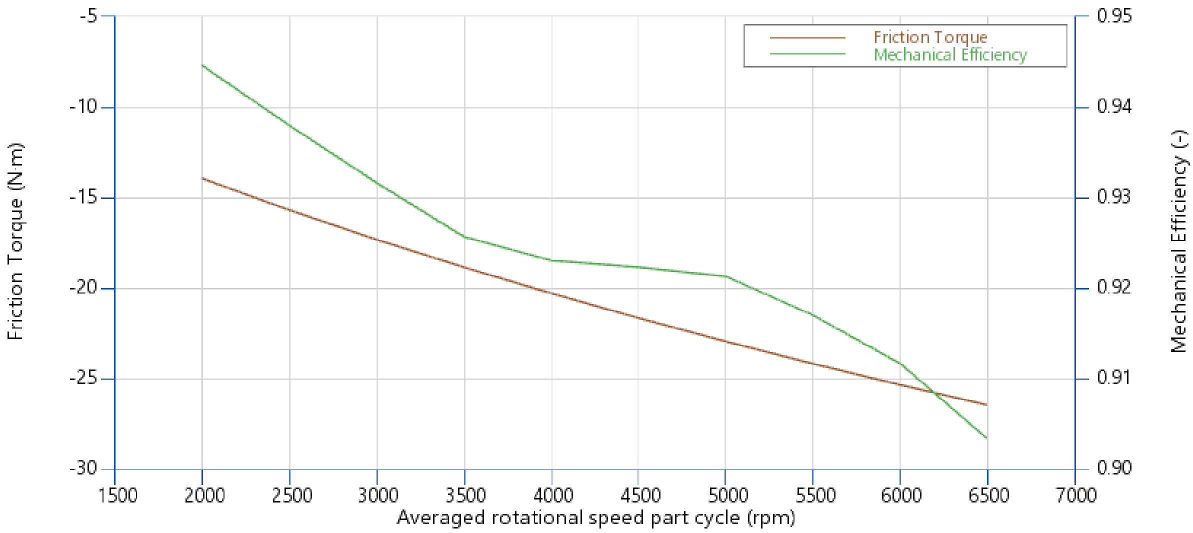
With these settings the following are the results.



3. Figure: Refined Engine Performance



4. Figure: Refined Engine BMEP & BSFC graph



5. Figure: Refined Engine Mechanical efficiency & Friction Torque

Figure 3 shows Effective torque (blue) and power (red) are plotted against engine speed. Torque holds flat low down, builds to a peak of 268 Nm at 5000 rpm, then drops off, while power keeps climbing to 225 hp (168 kW) at 6500 rpm. That split is the giveaway of a turbo engine working properly: torque tails off at the top but power still rises because the engine is spinning faster. It's the result that matters most, and the powerband is wide enough to be genuinely usable.

Figure 4 shows BMEP (purple) climbs from ~16.7 bar at low rpm to nearly 19 bar at 5000 rpm, while BSFC (orange) drops sharply from ~400 g/kWh down to ~220 g/kWh as rpm rises. BMEP is the key tuning metric — peak values near 19 bar confirm the boost and breathing mods load each cycle hard, well above naturally-aspirated levels (~12 bar). The very high BSFC at low rpm is expected here because the engine is off-boost and lightly loaded, so fuel is burned inefficiently; the figure falls to a healthy ~220 g/kWh once the engine is on boost and working in its efficient zone.

Figure 5 Mechanical efficiency (green) falls from ~0.945 at 2000 rpm to ~0.905 at 6500 rpm, while friction torque magnitude (red) grows from about −14 Nm to −27 Nm across the range. This quantifies how much indicated work is lost to friction before reaching the crank, partly explaining the high-rpm power taper. Both trends are textbook: friction rising with speed and mechanical efficiency dropping in step is exactly the expected behaviour.

Engine speed (rpm)	Effective Power (kW)	Effective Torque (N·m)	BMEP (bar)	BSFC (g/(kW·h))
2000	49.7	237.3	16.75	402.9
2500	62.0	237.0	16.72	401.3
3000	74.1	235.8	16.64	401.4
3500	86.0	234.5	16.55	402.0
4000	101.9	243.2	17.16	367.5
4500	121.0	256.8	18.12	307.8
5000	140.5	268.4	18.94	264.2
5500	153.7	266.9	18.83	242.4
6000	164.2	261.3	18.44	228.1
6500	168.1	247.0	17.43	223.4

4. Table: Engine Performance Results

2. Dyno data analysis and calibration review

The tuning work described in this chapter is not aimed at motorsport or competition use. The objective is street tuning, extracting more power and torque from the 1.8T engine while keeping the car fully usable and dependable in everyday driving. For that reason, the calibration was deliberately not pushed to the edge of what the engine could produce. A motorsport tune accepts narrow safety margins, shortened component life and constant supervision in exchange for the last few percent of output a street tune cannot. The engine must start reliably, run cleanly in all conditions, and survive years of use on pump fuel without close monitoring.

The priority throughout was therefore stability and safety rather than peak numbers. Power and torque were increased meaningfully over the stock calibration, but every change was made with knock margin, boost stability and combustion consistency taking precedence over chasing the highest possible figure on the dyno. This is why, as the results show, the run with the highest peak power was not the one chosen as the final calibration: the goal was a strong but dependable everyday tune, not a maximum-effort one

2.1 Dynamometer Setup and Preparation

The measurements in this chapter were carried out on a chassis (wheel) dynamometer rather than an engine dynamometer. The distinction matters: a chassis dyno measures power at the driven wheels with the engine still installed in the car, so the figures it produces already include the losses of the complete drivetrain. Because of friction and mechanical losses through the gearbox, driveshafts and tyres, the power measured at the wheels is typically around 15–20% lower than the power that would be measured directly at the engine's output [12.].

A chassis dynamometer works by placing the driven wheels on a set of rollers coupled to an absorption unit (commonly an eddy-current brake), which applies a controllable resistive load to simulate real-world driving resistance. The only quantities measured are the force at the load cell and the rotational speed at the roller encoder; everything else — torque, power, road speed — is calculated from these using the known roller radius and load-cell geometry [12.][13.].

Getting trustworthy data out of a chassis dyno depends far more on preparation than on the pull itself. The single most critical step is securing the car correctly. The vehicle is tied down so it cannot move fore, aft or sideways, both for safety and for measurement integrity. This is not a trivial detail: the way the car is strapped down actually changes the measured power, because the strap tension adds to the inertial and frictional losses the dyno sees. Straps that are inconsistent between runs will produce inconsistent numbers, so a repeatable tie-down procedure is essential if results are to be compared [13.][14.].

6. Figure: Preparation for the dyno run

Equally important is engine cooling. On the road the engine has a constant flow of air through the bay; on the rollers the car is stationary, so a dedicated cooling fan must be positioned in front of the vehicle to replicate that airflow and stop the engine and intake charge from heat-soaking during the pull. Without it, intake temperatures climb run-to-run and the data drifts. The dynamometer's own absorbers also need to be managed thermally — after a pull the car should be kept running at light load for a few minutes to let the eddy-current absorbers cool before they stop turning, which protects both the equipment and the consistency of back-to-back tests [13.][14.][15.].

Beyond these, proper instrumentation of the car is what separates a usable dataset from a meaningless one. For the combustion and ECU analysis that follows,

7. Figure: BDN Combustion Analyser setup

the car has to be fitted with the right sensors — cylinder-pressure transducer(s), a wideband lambda sensor, intake and coolant temperature channels, boost pressure, and a clean trigger/RPM reference — and every one of them has to be reading correctly and time-aligned before the first pull. A single mis-scaled or noisy sensor invalidates the whole run, and on a dyno you usually only find out afterwards. This is why the hardest part of dyno work is rarely the test itself; it is the preparation. Having every necessary tool and sensor in place, verified and calibrated beforehand, is what makes the difference, because a chassis dyno gives you a short, repeatable window in which everything must already be working.

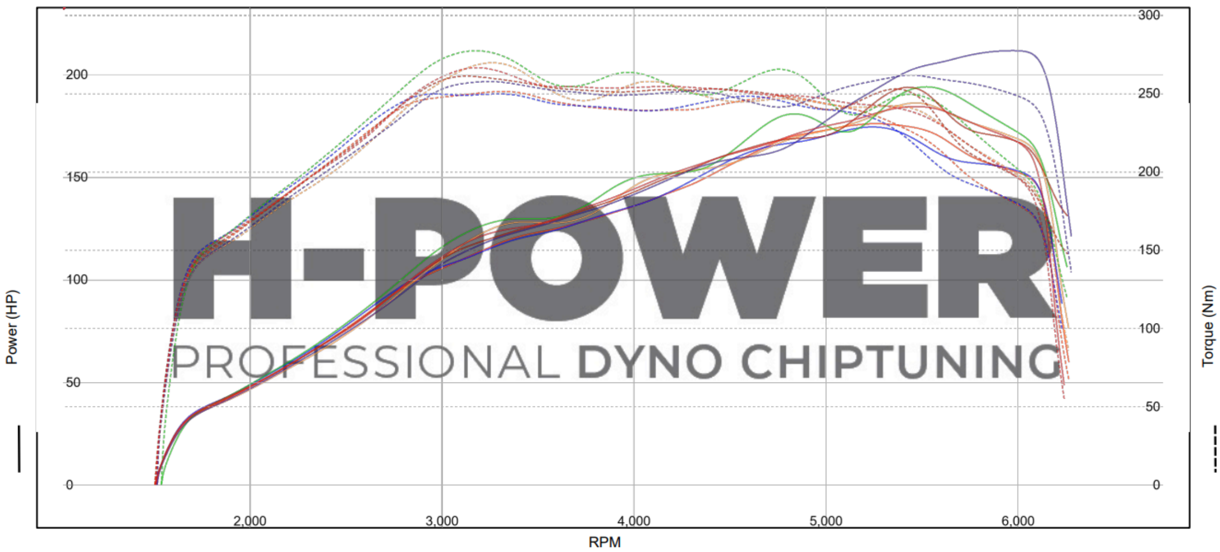
2.2 Measurement results

The measured vehicle was a Skoda Octavia 1.8T. It was measured on a chassis dynamometer in a single session at H-POWER Chiptuning, Budapest. Six pulls were carried out, labelled Run 2 to Run 7 on the dyno chart. After each run the results were reviewed in the Ignitron ECU software and, where available, the BDN combustion analyser, and the calibration was adjusted before the next pull. The aim throughout was to raise torque and power while keeping the engine reliable, so the changes were always made with combustion stability and knock in mind rather than chasing peak numbers blindly.

The main calibration levers were the boost control and the ignition/mixture maps. A target boost of 170 kPa absolute (1.7 bar) was set for the whole campaign; the actual manifold pressure was then shaped by adjusting the PID integrator maximum map, which limits how hard the controller can drive the wastegate. Ignition advance was tuned against the BDN knock-amplitude channel, and the lambda map was used to add torque in the lower rpm range where ignition advance alone was running out of room.

Test conditions: ambient temperature 25–27 °C, SAE correction factor 1.007–1.010 across all runs, so the figures are correction-normalised and directly comparable.

The file and run naming were inconsistent this session and this is something which should be improved for the next dyno run because it can be confusing for the reader and can cause communication problems, better preparation and file mapping is necessary.



	Engine power (HP)	RPM	Torque	RPM	Temp (°C)	SAE
Run 7	184.4	5475	266.5	3175	26.1	1.008
Run 6	211.7	5975	261.7	5450	26.4	1.009
Run 5	193.9	5425	261.1	3125	26.6	1.010

	Engine power (HP)	RPM	Torque	RPM	Temp (°C)	SAE
Run 4	194.1	5525	277.4	3175	26.4	1.010
Run 3	186.2	5475	269.7	3275	25.6	1.008
Run 2	174.5	5250	249.7	2975	25.3	1.007

8. Figure: Dyno Measurement results

Run 2 - Base Run

The first pull was a pure reference with the stock calibration, to establish a baseline for the later adjustments. It produced 174.5 HP and a peak torque of 249.7 Nm. No BDN data exists for this run because the combustion sensor was not reading, so it was assessed from the Ignitron log alone. The log showed boost building to around 150–160 kPa by ~3000 rpm but staying below the 170 kPa target for the whole run, then falling away after ~5300 rpm. This confirmed the engine ran cleanly but never reached target boost, which set the direction for the following runs.

Run 3 - Base Repeat

A repeat of the baseline, again without BDN data (sensor still down). Output rose to 186.2 HP and 269.7 Nm. The manifold pressure built faster and higher at low rpm than in Run 2, overshooting the target slightly in the mid-range (roughly 10 kPa overboost around 3000–3600 rpm). This showed the boost could be brought up to target, but the regulation around it was not yet stable — overboost waviness appeared.

Run 4 - PID integrator tuning

Here the integrator maximum map was adjusted to improve target-holding. This run gave the strongest headline figures of the session: 194.1 HP and the highest torque of all, 277.4 Nm. Boost build-up was present but the mid-range manifold pressure was still wavy rather than flat, and the BDN IMEP channel mirrored that unevenness, confirming the boost regulation still needed refinement.

Run 5 - Ignition advance + Lambda change

With the integrator map refined further, ignition advance and the lambda map were now also tuned: a leaner mixture below 4000 rpm to gain torque, and retarded timing above 4000 rpm where the BDN knock-amplitude channel showed rising knock. Power was effectively unchanged at 193.9 HP and torque settled to 261.1 Nm — slightly down on Run 4's peak, but the trade was deliberate, buying knock margin and a cleaner mixture rather than chasing the number.

Run 6 – Ignition refinement

The integrator map was extended into the higher rpm range for steadier boost at the top end, with continued ignition work. This run recorded the highest peak power of the whole session, 211.7 HP, with 261.7 Nm of torque. Boost tracked the target steadily up to about 5500 rpm before tapering smoothly, with much less of the waviness seen earlier, and the IMEP trace was correspondingly smoother.

Run 7 -Ignition + turbo adjustment

The final pull pushed the integrator maximum map higher again to hold target boost at high rpm, with further ignition refinement. The headline figures were the lowest of the tuned runs — 184.4 HP and 266.5 Nm — but this run gave the most stable behaviour overall: boost held closest to target across the range, the IMEP trace was the most even, and knock, while still present at high rpm, was reduced compared with earlier runs.

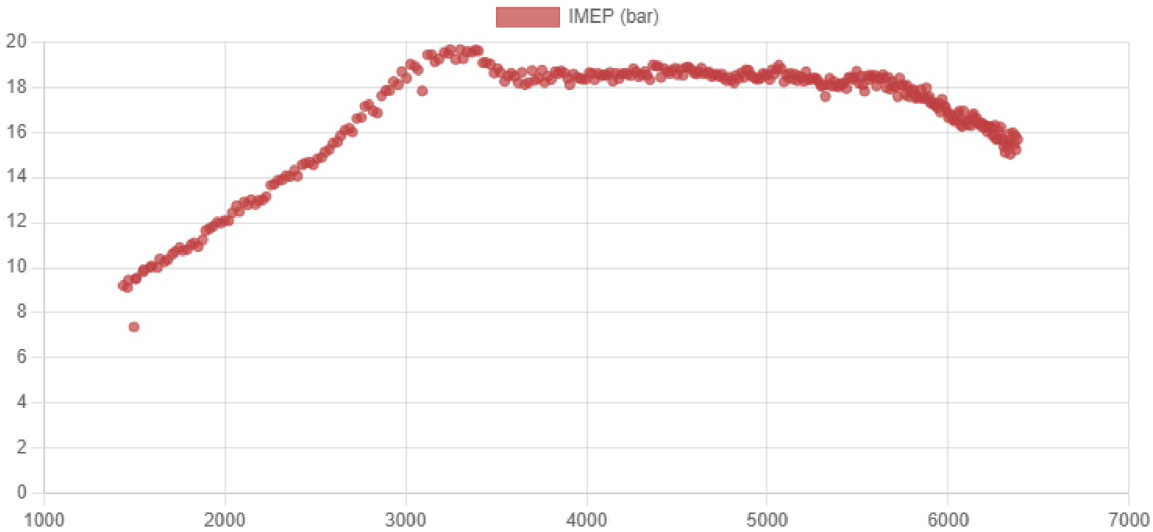
Overall assessment

Across the session, the runs that produced the biggest headline numbers (Run 6 for power, run 4 for torque) were not the same as the run judged best overall. Run 7, despite its lower peak power, gave the most balanced result in terms of boost build-up, target-holding and combustion stability, which is why it is the most suitable run to use as the reference for calibrating the simulation model, and for the everyday usage. This illustrates a point worth making explicitly: on a real engine the best calibration is the one that combines good output with stable, knock-safe combustion, not simply the one with the highest peak on the chart.

Run	Calibration change	Peak Power (HP)	@ RPM	Peak Torque (Nm)	@ RPM	Δ Power vs base	Δ Torque vs base
2	Base run	174.5	5250	249.7	2975	-	-
3	Base repeat	186.2	5475	269.7	3275	+11.7 (+6.7%)	+20.0 (+8.0%)
4	PID integrator max map	194.1	5525	277.4	3175	+19.6 (+11.2%)	+27.7 (+11.1%)
5	Ign. advance + lambda	193.9	5425	261.1	3125	+19.4 (+11.1%)	+11.4 (+4.6%)
6	Ign. upper-range pull-back	211.7	5975	261.7	5450	+37.2 (+21.3%)	+12.0 (+4.8%)
7	Ign. + turbo	184.4	5475	266.5	3175	+9.9 (+5.7%)	+16.8 (+6.7%)

5. Table: Run-by-run calibration changes and peak results

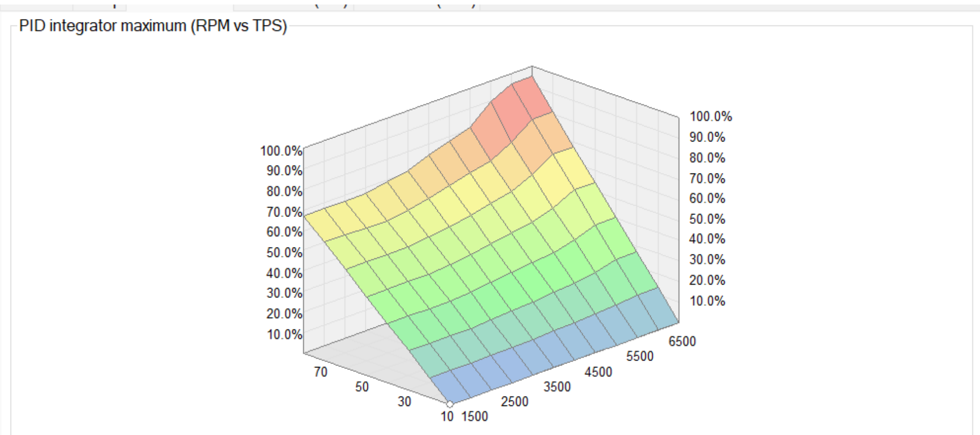
The following figure shows IMEP (indicated mean effective pressure) against engine speed from the BDN combustion analyser (Run 7). IMEP measures the useful combustion work done in the cylinder each cycle, making it a key indicator of how cleanly and consistently the engine is burning across the rev range. The trace peaks at around 19.5 bar near 3200 rpm, holds a stable 18–18.5 bar plateau through the mid-range, then tapers to 15–16 bar by 6300 rpm. This is a healthy shape, showing consistent combustion load with no instability in the usable range, and the peak agrees closely with the ~19 bar BMEP from the simulation — confirming the model and the real engine are consistent. (The end-of-pull spin-down cycles, where IMEP collapses to near zero, have been excluded from the figure.)



9. Figure: IMEP vs RPM (BDN comustion analysis)

Figure 10 shows the final PID integrator maximum map, plotted as a function of engine speed (rpm) and throttle position (TPS). This map sets the upper limit of the boost controller's integral term — in effect, how much authority the controller has to drive the wastegate closed to chase the target boost in each operating region. Higher values let the system build and hold more boost; lower values restrain it.

The shape reflects the calibration logic followed throughout the measurements. At low throttle the values stay small across the whole rev range, keeping boost gentle and the car docile in part-load driving. At high throttle the values rise with engine speed, reaching their maximum in the upper rpm range, which is precisely where the earlier runs showed the boost falling away from target. Increasing the integrator ceiling here was the change that allowed the final run to hold the 170 kPa target much more steadily at high rpm. This map is therefore the practical result of the iterative tuning process: each run's boost behaviour fed back into reshaping these values, and the version shown represents the setup that gave



%	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000
80	67.0	67.0	67.0	67.0	69.0	71.0	75.0	78.0	81.0	90.0	95.0	95.0
70	57.4	57.4	57.4	57.4	59.1	60.8	63.4	66.0	68.5	73.7	81.4	81.4
60	47.8	47.8	47.8	47.8	49.2	50.7	52.8	55.0	57.1	61.4	67.8	67.8
50	38.2	38.2	38.2	38.2	39.4	40.5	42.2	44.0	45.7	49.1	54.2	54.2
40	28.7	28.7	28.7	28.7	29.5	30.4	31.7	33.0	34.2	36.8	40.7	40.7
30	19.1	19.1	19.1	19.1	19.7	20.2	21.1	22.0	22.8	24.5	27.1	27.1
20	9.5	9.5	9.5	9.5	9.8	10.1	10.5	11.0	11.4	12.2	13.5	13.5
10	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

10. Figure: Final PID Integrator Maximum Map

the most stable, target-holding boost while keeping the engine safe at part load.

3. Correlation analysis

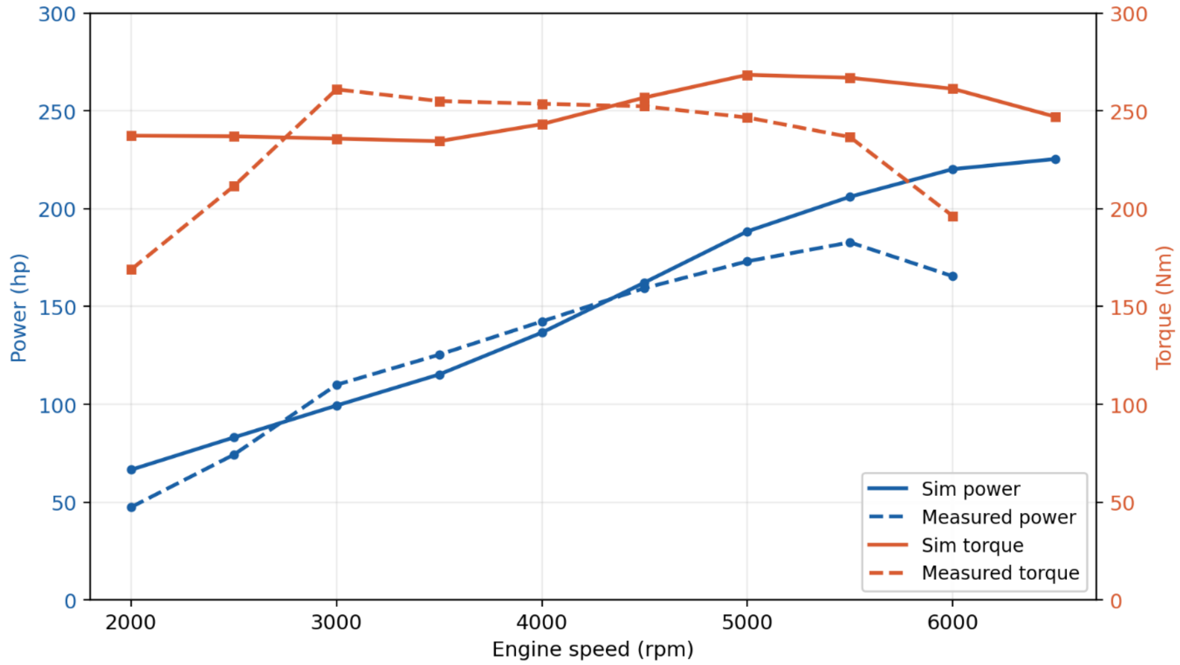
Correlation analysis is the step that gives the simulation its credibility. A model is only useful if its outputs can be trusted to represent the real engine, and the only way to establish that trust is to compare the simulated results directly against measured data from the test bench. Without this step, the simulation remains an unvalidated prediction; with it, the model becomes a verified tool that can be used with confidence for further development work, such as exploring modifications without the time and cost of physical testing [16.].

The aim of this chapter is therefore to quantify how closely the simulation matches the measured dyno results, with a target difference of below 5% in the key performance figures. Where the deviation exceeds this threshold, the differences are identified and a root-cause analysis is carried out to explain them, whether they stem from modelling assumptions, input data, or unrepresented physical effects. Based on that analysis, refinements are proposed and applied to the simulation, and their effect on the correlation is reviewed. The reference used for this comparison is the final H-POWER dyno run, which was selected earlier as the most stable and representative measurement of the engine's behaviour.

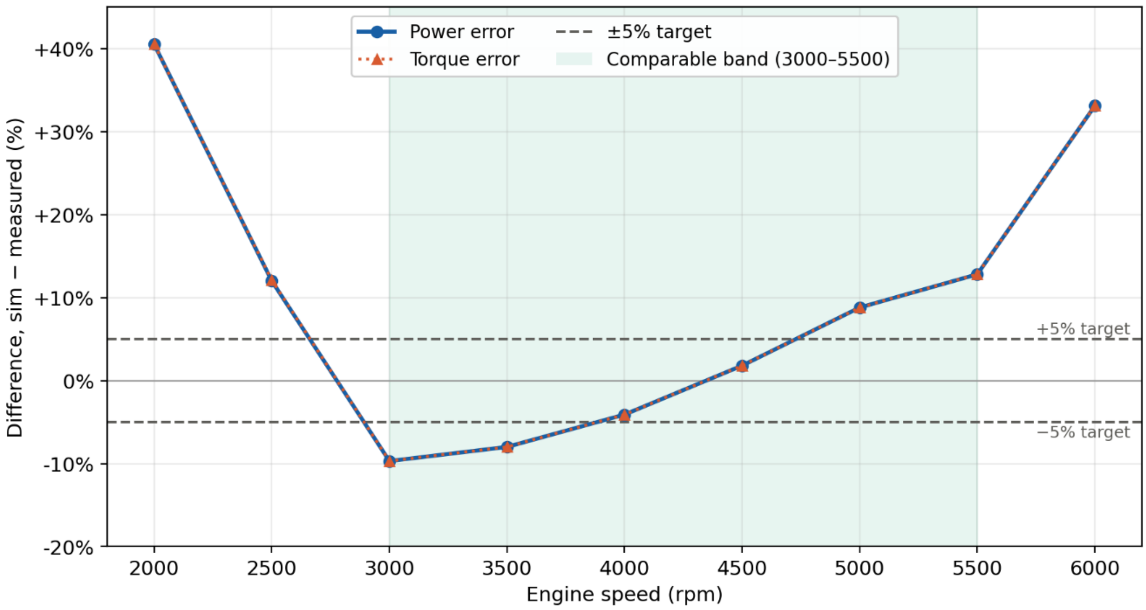
3.1 Comparison of Simulation and Measurement

Figure 11 shows the simulation result plotted against the measured result (Run 7).

Both the Ignitron (ECU) logs and the H-POWER chassis-dyno results were available, but the Ignitron data reports brake power and torque at the engine output — the same quantity the simulation computes — whereas the chassis dyno measures wheel power with an estimated drivetrain loss added back. The Ignitron dataset is therefore the better reference for the correlation, and reassuringly both sources peak at almost the same power (around 210–211 hp), confirming they broadly agree.



11. Figure: Simulated and Measured engine power and torque



12. Figure: Correlation error between simulation and measurement

3.2 Root-Cause Analysis of the Correlation

The comparison between the simulation and the final dyno run (Run 7) shows three distinct regions, and they have to be treated separately because not all of them represent a fair comparison between model and engine.

In the mid-range, between roughly 3000 and 5500 rpm, both the engine and the model are operating at full load, so this is the only band where the two can be compared on equal terms. Here the agreement is reasonable but not yet within target: the mean absolute difference is about 7.5% for both power and torque, with the best agreement around 4000–4750 rpm where the error falls inside the ±5% band, and the largest deviation of about 12.8% at the edges. The simulation slightly under-predicts at the lower end of this band (around −8 to −10%) and over-predicts towards the top (up to +12.8%). This crossover pattern, rather than a constant offset, points to the cause being the shape of the modelled torque curve rather than a simple scaling error. The most likely contributor is the combustion model: a fixed combustion phasing in the simulation cannot reproduce the way the real engine's ignition timing and effective burn shift across the rev range, so the model puts the torque emphasis at a slightly different engine speed than the real engine does.

The boost model is a second contributor — the simulation regulates to a clean target pressure ratio, whereas the measured engine showed the boost building and tapering less ideally, which shifts where the torque peaks.

At low rpm, below about 3000 rpm, the difference grows very large (up to about 40%), but this is not a model error. The dyno data here comes from the start of the pull, where the engine is still loading up and is not yet at full throttle and full boost, while the simulation computes a fully loaded steady-state point at every speed. The two are therefore simply not describing the same operating condition, and the apparent error is a measurement artefact of the transient start of the run rather than a fault in the model.

At high rpm, above about 5500 rpm, the difference again rises sharply (over 30% by 6000 rpm). This too is largely an artefact of the measurement: towards the end of the pull the measured power collapses as the run is brought to an end and the boost tapers, and the dyno record stops around 6250 rpm, so there is no measured data at 6500 rpm to compare against at all. The simulation, having no such cut-off, continues to predict rising power to the top of the range. Some genuine model contribution is likely mixed in here — the simulation does not capture the full extent of the real engine's high-rpm boost fall-off and increasing friction — but the bulk of the gap is the end-of-run behaviour of the test rather than a modelling defect.

A point worth noting is that the percentage errors for power and torque are almost identical at every speed. This is expected: at a given engine speed, power is simply torque multiplied by rotational speed, so when the simulated and measured values are compared at the same rpm, the power ratio equals the torque ratio and the two percentage errors must coincide. It is a consequence of the definition rather than an independent result, and it confirms the comparison was carried out consistently.

3.3 Proposed refinements

The single change expected to bring the largest improvement is a more detailed turbocharger model. The current simulation regulates boost towards an idealised target pressure ratio, which cannot fully reproduce how the real K03s turbo builds and tapers boost across the rev range — and since boost shapes the whole torque curve, this is the most likely source of the crossover error seen in the comparable band. A properly parameterised turbo model would need the compressor and turbine maps describing the unit's actual flow and efficiency behaviour. These can be obtained either from turbocharger test-bench measurements or from the manufacturer's datasheets, neither of which was available for this project; this is therefore the clearest direction for future work, where access to that data would allow the boost behaviour to be modelled far more accurately rather than approximated.

Alongside this, two further refinements would improve the correlation: introducing speed-dependent combustion phasing, so the modelled torque curve is shaped correctly across the rev range instead of peaking at a fixed point and improving the high-speed friction representation so the predicted top-end power rolls off more realistically. Both are achievable with the data already available and would complement the turbo model, but the turbocharger remains the dominant factor and the area where better input data would yield the greatest gain in accuracy.

4. Benchmarking and Impact Analysis

This chapter places the engine in the context of comparable tuned 1.8T 20V builds, comparing its performance against documented aftermarket results and the modifications behind them, and then identifies which design and calibration variables had the greatest influence on the outcome. As this engine is a street-oriented build rather than a competition unit, the most relevant benchmarks are other road-going and tuned transverse 1.8T engines rather than dedicated motorsport powertrains.

4.1 Benchmarking against tuned 1.8T builds

The 1.8T 20V is one of the most extensively tuned engines in the VAG range, and the aftermarket has well-established "stages," each defined by a specific set of modifications and a typical resulting output. This allows the engine to be benchmarked not just against factory figures but against documented tuned builds and the changes that produced them. The relevant ladder for a K03S-based engine like this project is:

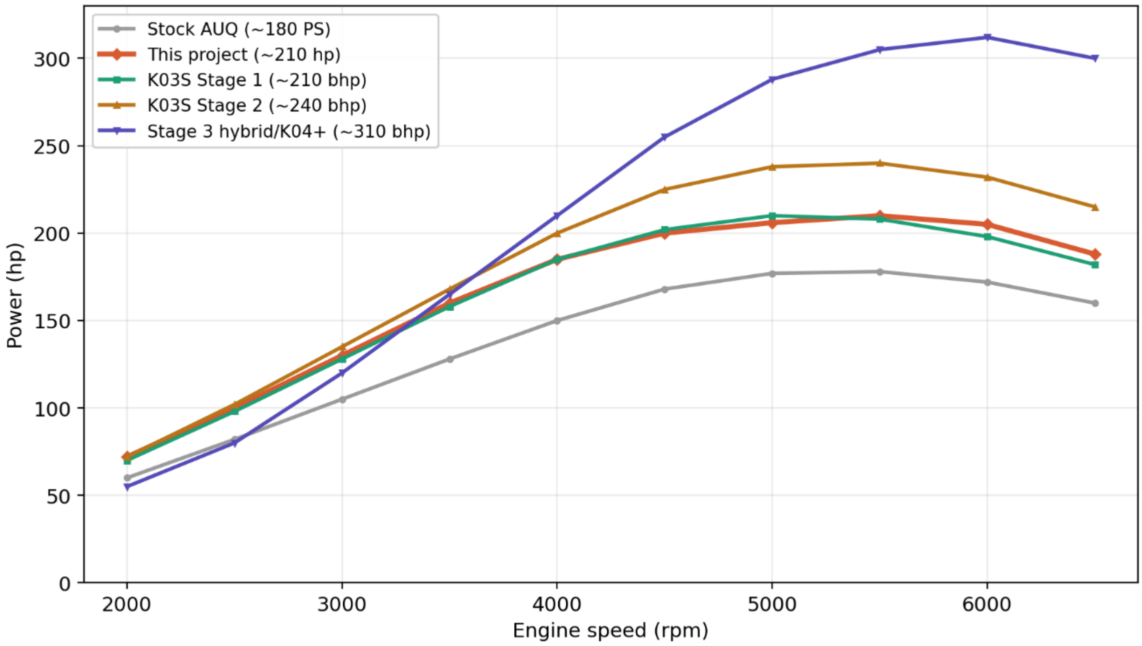
Stock AUQ — 180 Hp and roughly 235 Nm on the factory K03S turbo and factory map.

K03S Stage 1 — typically around 210 Hp and 330 Nm, achieved through an ECU remap alone that raises boost and optimises ignition and fuelling, optionally with a panel filter or turbo intake pipe, and no turbo change. This is the closest benchmark to this project, as it relies on the same calibration-led approach on the same turbo.

K03S Stage 2 — typically 235–250 Hp with 390 Nm, adding a 3″ decat downpipe, free-flowing exhaust, front-mount intercooler and uprated intake. The gains come from reducing exhaust backpressure and intake temperatures so the same turbo can be driven harder, often with a stronger wastegate actuator to hold boost into the mid-range.

Stage 3 / hybrid-turbo builds — typically 260–350+ hp, achieved by moving beyond the factory turbo entirely with a hybrid K04 or larger frame turbo, larger injectors, an uprated fuel pump and supporting Stage 2 hardware. Above roughly 300 hp the factory connecting rods become a liability, so forged internals are expected. This is a fundamentally different class of build, changing the engine's core hardware rather than optimising what is already there.

Positioned against this ladder, the project is firmly a calibration-led K03S build in the Stage 1 mould: the modifications (sport catalyst, 76 mm exhaust, front-mount intercooler, RS3 coils) combined with boost and ignition tuning, with no turbo change and stock internals. Its ~210 hp output and ~19 bar BMEP place it squarely at the Stage 1 level, exactly where its hardware would predict.



13. Figure: Power curve benchmarking

Plotted against the tuning ladder, the project engine sits almost exactly on the K03S Stage 1 line. It reaches the ~210 hp typical of a Stage 1 K03S engine, with the same mid-range-strong, high-rpm-tapering shape that is the signature of the small K03S turbo, and it clearly outperforms the stock 180 PS curve across the whole rev range while staying below the Stage 2 and Stage 3 lines.

A recurring theme in the 1.8T tuning literature is that the small K03S turbo runs out of breath at the top end: it produces a strong mid-range but boost, and therefore power, tapers above roughly 5000–5500 rpm because the turbine cannot supply more flow. The project's measured curve shows exactly this signature, confirming the engine behaves like a correctly tuned K03S unit and that the turbo, not the calibration, is the ceiling on top-end power. The Stage 3 curve illustrates the contrast: a larger turbo spools later — sitting below the smaller-turbo curves at low rpm — but once lit it climbs hardest and holds power to the rev limit instead of tapering, which is the opposite of the project's behaviour.

It is worth emphasising that this result represents a deliberately conservative use of the setup rather than its limit. The dyno session showed that earlier runs produced higher peak figures than the calibration ultimately chosen, so the engine demonstrated more potential than the final map extracts. The owner specifically opted for the more modest, stable calibration for everyday driving, prioritising reliability, knock margin and consistent boost over the highest possible number. The engine could therefore sit higher toward the Stage 2 line, but was intentionally kept at a safe, repeatable Stage 1-level tune because it is a daily-driven streetcar rather than a competition engine chasing a peak dyno figure. The engine benchmarks at Stage 1 by choice, not by limitation, and the measured curve understates what the same hardware could deliver if pushed.

4.2 Which variables had the largest influence

The tuning ladder itself reveals which variables matter most, because each stage is defined by what it changes:

Boost (calibration) is the dominant lever. The entire Stage 1 gain — roughly +30 hp and around 30% over stock — comes from a remap that raises and holds boost, with no hardware change. This is the single biggest step on the ladder and matches the project's own finding that the boost (PID integrator) map produced the largest gain.

Ignition timing (calibration) is the variable that converts the additional air into torque safely. It is knock-limited, which is why good fuel is stressed throughout the literature and why the project's ignition tuning was deliberately bounded by knock margin rather than maximised.

Intake and exhaust flow (design) is the defining change of Stage 2. Its effect is real but secondary: it does not make power by itself, it removes the restrictions that allow boost to be sustained, lowering intake temperatures and backpressure.

This mirrors the project's conclusion that the breathing modifications were enablers rather than primary contributors.

Turbo capacity (design) is the hard ceiling. The fact that progression to Stage 3 requires a turbo change — no amount of further boost or ignition work on the K03S can reach those numbers — shows that once boost and flow are optimised, the turbocharger becomes the single limiting design variable. The project's high-rpm taper is the K03S reaching that limit; the Stage 3 curve holding power to redline is what removing the limit looks like.

In summary, both the literature and the project's own results agree: on a K03S 1.8T, calibration variables — boost first, ignition second — deliver the majority of the gains, breathing modifications enable those gains to be held, and the turbocharger sets the ultimate limit. The project's headroom reinforces this, since the engine had more to give on boost and ignition, meaning it was the calibration choice rather than the hardware that set the final output.

5. Technical Recommendations

This chapter draws together the unusual patterns and limitations observed across the simulation, the dyno measurements and the correlation, and offers practical recommendations for improvement, with particular attention to changes that deliver worthwhile gains at low cost.

5.1 Observed patterns and limitations

Several recurring features in the results point to where the engine and the model are limited:

The clearest pattern is the high-rpm power taper. Both the measured curve and the literature comparison show power falling away above roughly 5500 rpm, which is the K03S turbo reaching its flow limit rather than a calibration fault. This is the single biggest limitation on top-end output and is inherent to the small turbo.

The boost trace showed waviness and overboost in the earlier runs before the PID integrator map was refined. While the final calibration largely resolved this, it highlights that the boost control is sensitive and that the stock wastegate actuator struggles to hold boost cleanly in the mid-range as load increases.

The correlation analysis revealed the simulation deviating from measurement by about 7.5% in the comparable band, above the 5% target. The crossover shape of the error — under-prediction low down, over-prediction up top — pointed to the idealised turbocharger and fixed combustion phasing in the model rather than a single scaling error.

The low-rpm BSFC was high, consistent with the engine being off-boost and lightly loaded there, and the BDN data contained the expected end-of-run artefacts that had to be filtered. These are measurement characteristics rather than engine faults, but they limit how much of the data is directly usable.

5.2 Cost-effective recommendations

Focusing first on changes that offer the best return for least cost:
The most cost-effective single improvement is a stronger wastegate actuator (or correctly set boost-control hardware). This is an inexpensive part that directly addresses the mid-range boost waviness seen in the data, allowing the existing turbo to hold target boost more cleanly without any change to the turbo itself. It improves the usable torque curve and the consistency of the tune for very little outlay.

Ensuring the supporting hardware is in good order — a strong diverter valve, healthy N75 valve, leak-free charge pipework and good-quality fuel — is effectively free or low-cost and is repeatedly stressed in the tuning literature as the difference between a clean tune and a compromised one. Several of the early-run inconsistencies could stem from exactly these items, so verifying them is a sensible first step.

Improving intake-charge cooling and flow is the next tier: the project already runs a front-mount intercooler but ensuring intake air temperatures stay low across repeated pulls protects ignition timing and therefore power, since the tune is knock-limited. A high-flow turbo intake pipe and panel filter are modest-cost items that the literature associates with a few horsepower and improved spool, without requiring a remap.

On the calibration side, the engine has demonstrated headroom that was deliberately left unused for everyday reliability. If a slightly more aggressive but still safe map were desired, the cheapest "gain" available is simply recovering some of that margin through careful ignition and boost work within the K03S's capability — no hardware cost, only tuning time, and bounded by knock margin.

For the simulation specifically, the most cost-effective refinement is introducing speed-dependent combustion phasing, which requires no new data and would directly reduce the crossover error in the comparable band. A more accurate turbocharger model would give the largest improvement overall but depends on compressor and turbine maps that are not freely available, so it is best treated as future work rather than an immediate, low-cost step.

Finally, it must be recognised that any substantial increase in peak power beyond the current Stage 1-level output requires a turbo change (Stage 2/3), which is a significant cost and falls outside the "marginal gains" scope. The recommendations above are therefore deliberately focused on extracting the most from the existing K03S hardware — better boost control, healthy supporting components, lower intake temperatures and improved calibration accuracy — which together represent the genuinely cost-effective path to a better, more consistent result.

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